

THE EASTERN AFRICA STATISTICAL TRAINING CENTRE



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DEPARTMENT OF OFFICIAL STATISTICS

ASSESSMENT PLAN FOR **HIGHER DIPLOMA IN DATA SCIENCE -NTA LEVEL 7**

1. INTRODUCTION

- MODULE CODE: **DSU07315**
- MODULE NAME: **BASICS OF BIG DATA**
- TOTAL CREDITS: **08**
- COVERAGE WEEKS: **16 WEEKS**

2. PRINCIPAL LEARNING OUTCOMES AND ENABLING OUTCOMES

Principal learning outcomes	Enabling learning outcomes
4.0	4.1

3. ENABLING AND SUB-ENABLING LEARNING OUTCOMES

Enabling learning outcomes	Sub-Enabling learning outcomes	
4.1	4.1.1	Describe the concepts of Big Data
	4.1.2	Describe the architecture of Big Data
	4.1.3	Apply big data processing models to summarize and explore big data
	4.1.4	Describe big data analytics techniques

4. TABLE OF SPECIFICATION

The table below clarifies the sub-enabling learning outcomes and their related tasks will be assessed

Sub-enabling and Related Task	COMPETENCES TO BE ASSESSED				CATEGORIES OF ASSESSMENT	
	Knowledge	Skills	Understanding	Wide Attributes	CA	SE
Sub-enabling learning outcomes						
4.1.1 Describe the concepts of Big Data						
Related task						
(a) Define the Big Data concept	√		√		√	
(b) Explain the 4V's/5V's of Big Data			√		√	
(c) Explain the challenges in Big Data			√	√		√
4.1.2 Describe the architecture of Big Data						
(a) Define Big Data Processing Cycle	√		√		√	
(b) Explain the basic phases of Big Data Processing Cycle.			√		√	
(c) Explain the architectural layers in Big data		√	√			√
(d) Explain the data storage in Big data. E.g		√	√			√
4.1.3 Apply big data processing models to summarize and explore big data						
(a) Define big data processing models	√		√		√	
(b) Differentiate between batch processing and real-time processing models	√	√	√			√

Sub-enabling and Related Task	COMPETENCES TO BE ASSESSED				CATEGORIES OF ASSESSMENT	
	Knowledge	Skills	Understanding	Wide Attributes	CA	SE
(c) Explain some examples of big data processing models	√		√			√
(d) Use Big data processing models to summarize and explore big data	√	√	√		√	
4.1.4 Describe big data analytics techniques						
(a) Define the concept of big data analytics techniques	√		√		√	
(b) Explain some big data analytics techniques. E.g -Network Analysis -Machine learning -Sentiment Analysis -Cluster Analysis	√		√		√	
(c) Explain the applications of Big Data analytic techniques.	√		√			√
(d) Explain the challenges of Big data analytic techniques.	√		√			√

5.0. INSTRUCTIONS

Assessment Methods

Methods of assessment for CA and SE in this module are as follows:

Continuous Assessments (CA) **40%**

The continuous assessment will consist of:

Written test 15%,

One Practical Assignment 15%

Assignments 10%

The details of the continuous assessment components are as follows:

Written Test

There shall be one written test

The duration of the test shall be 1 to 1.30 hours

The test will carry **15%**.

The test will cover two (2) SEOs and four (4) RTs with a weight of 15% as elaborated below:

4.1.1 SEO three (2) RTs which are; a and b

4.1.2 SEO four (2) RTs which are; a and b

Practical Assignment will cover one (1) SEO and seven (2) RTs with a weight of 15%

The distribution will be as follows:

4.1.3 SEO one (1) RT which are: a and d

Assignment:

There shall be two assignments; each will carry 5% to make a **total of 10%**.

The assignment one will be derived from 4.1.4 SEO one RT which is a

The assignment two will be derived from 4.1.4 SEO one RT which is b

Semester Examination (SE)

End of semester examination will cover four (4) SEOs and seven (7) RTs with **weight of 60%**

The duration for the SE shall be 3 hours

The distribution will be as follows;

4.1.1 SEO one (1) RT which is c

4.1.2 SEO two (2) RT which are c and d

4.1.3 SEO one (1) RT which are b and c

4.1.4 SEO three (3) RT which are; c and d